Which Inputs Carry a Style? Custom Style Transfer and Its Cost in Real-Time Diffusion on Consumer Hardware

Submission 8

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Abstract

This paper presents a case study on real-time style transfer with a self-trained LoRA. I trained a rank-16 adapter on 150 renders of my own 3D artwork and ran it in StreamDiffusionTD on an 8 GB laptop GPU, asking which properties an input must retain for the style to reach the output. Eight inputs, from the original Cycles render down to grey noise, were streamed with and without the adapter and rated against four criteria fixed in advance. The adapter improved every condition, but not in the expected order. A black-and-white mask outscored a fully shaded render, and a low-poly render matched the original. Figure-ground contrast mattered more than input reduction, and brightness shaped feature placement. The custom adapter cost nothing in speed or memory, at 7.5–8.0 fps with and without it. One denoising step raised it to 11.8 fps; live input on the same GPU lowered it to 6.6.

1. Introduction

1.1. Motivation

The motivation for this project originates from an artwork I created as branding for a music festival. Produced in Blender and rendered with the Cycles path-tracing engine, it consists of abstract organic shapes generated through procedural sculpting. The main form alone carries about 916,000 triangles, and one 1920×1080 frame took about 30 seconds at 1000 samples, roughly twelve hours for a minute of animation. Long sequences were therefore impractical, and live use was difficult.

This limitation motivated me to test an alternative workflow, training a custom LoRA adapter on the artwork's visual style and driving a real-time img2img pipeline with a simplified, faster-to-render representation of the original artwork. The adapter could then be used to reintroduce stylistic properties such as material appearance and lighting during inference.

How much visual information an input must retain for the learned style to appear is unclear. In a real-time img2img pipeline, the input constrains the output structure, while the style is contributed by the adapter, so simplifying the input may either weaken style expression or leave more room for it to emerge. This question also extends to inputs that no longer depict the original scene, such as animated noise or abstract structures built with TouchDesigner networks. In addition, personalisation may introduce computational costs that matter on consumer hardware.

Both questions touch on existing research, but from opposite ends. The real-time work I reviewed is built around throughput, and the personalisation work evaluates style at full step counts and from a prompt rather than from a reduced image (Section 2). This study sits where the two meet, with a personally trained style, few denoising steps, and an input produced quickly or live. It therefore asks the following questions:

Research Question. Which properties of the input image determine whether a custom-trained style appears in the output of a real-time diffusion pipeline, and what does personalisation cost on consumer hardware?

SQ1 (Fidelity). Which characteristics of the input determine style fidelity when the input is varied along a cascade of decreasing similarity to the training data?

SQ2 (Cost). What does personalisation cost in throughput and memory—for the fused adapter, for the denoising step budget, and once the input is generated on the same GPU?

Contributions. (i) An eight-condition comparison of input representations, each run with and without a self-trained style adapter. (ii) The finding that similarity to the training data does not predict style fidelity, with a hypothesis from the eight cases in which figure-ground contrast orders the results and brightness shapes which learned elements appear. (iii) A cost profile in which the adapter shows no measurable difference in throughput or memory, the denoising step budget is the effective lever on throughput, and the binding constraint is not memory but contention for the GPU with the image source, about 15 %.

I write in the first person throughout, since I am at once the artist whose style is transferred, the operator of the pipeline, and the sole rater.

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2. Related Work

Running diffusion at interactive rates rests on a stack of accelerations, all of them used in this study. Latent diffusion models denoise in a compressed latent space rather than at pixel resolution [RBL*22]. Latent Consistency Models reduce the number of denoising steps required [LTH*23], and LCM-LoRA delivers that acceleration as an inference-time adapter [LTP*23]. StreamDiffusion assembles these into a streaming pipeline with batching, reused guidance, and optional frame skipping [KXH*23]; its TouchDesigner integration provides the deployment context of this study. What this work does not report is what the pipeline produces when the input differs substantially from the data used for personalisation.

That question is closely tied to how image-to-image diffusion treats its input. The source image is encoded into the latent space and partially replaced with noise before denoising; how much is added determines how closely the output remains tied to it [MHS*22]. The input therefore acts as a spatial and semantic constraint on generation, yet it remains unclear which of its visual properties must be preserved when it is simplified for real-time use. The style side of the problem is commonly addressed through Low-Rank Adaptation (LoRA), which trains a small number of additional parameters instead of updating the full model [HSW*22], making artist-specific adaptation feasible on consumer hardware.

Artist-led work treats a small, self-curated set of works not as a data shortage, but as a deliberate source of creative influence. Vigliensoni et al. argue that small datasets let practitioners steer a model by changing a few examples [VPF22]. Grimes and Harrison document this approach with LoRA fine-tuning on a single artist's corpus for text-to-image generation, but note the absence of quantitative evaluation as a limitation [GH26].

What remains unsettled is how much the input contributes. Akten et al. show that darker inputs lead their system to draw on darker learned features, while brighter areas draw in brighter ones, so the input selects which learned features appear [AFG19]. However, their work describes rather than measures this effect, and uses a conditional GAN with a camera feed rather than a few-step diffusion pipeline with a self-trained adapter. In the work I reviewed the input's contribution is therefore either described without measurement or does not arise, since prompt-based generation has no image input. To the best of my knowledge, the combination of a self-trained adapter, few-step streaming inference, and deliberately simplified input has received little systematic evaluation.

3. System Overview and Setup

The study comprises four recorded phases, in which eight input conditions were streamed through the pipeline described below, with and without the style adapter.

3.1. Pipeline

All runs used Stable Diffusion 1.5 ("v1-5-pruned-emaonly-fp16") inside StreamDiffusionTD 0.3.1 [Hin25], built on StreamDiffusion [cum24], running in TouchDesigner 2025.33230. Following the combination Luo et al. describe [LTP*23], two adapters were merged at startup, LCM-LoRA (slot 0, weight 1.0) for few-step sampling and a custom style adapter (slot 1, weight 0.9). Setting the first slot to weight zero also removed the effect of the second, so slot order is part of the configuration. The pipeline ran at 512×512 with TAESD, acceleration "none", "cfg_type: self", guidance scale 1.0, and the stochastic similarity filter disabled (no frame skipping). Hardware: Windows 11 laptop, NVIDIA RTX 5070 Laptop GPU (8 GB, Blackwell), 64 GB RAM. TensorRT, NVIDIA's toolkit for accelerating inference, was not used in this study; production support for this GPU generation is listed only from a later component release [Hin25]. ControlNet, IP Adapter and StreamV2V all require it in v0.3.1 and were therefore outside the scope of these runs.



Figure 1: *Twelve of the 150 Cycles renders used to train the style adapter.*

3.2. Style Adapter Training

The adapter was trained on 150 frames at 512×512 from the source artwork, rendered in Blender with Cycles (Fig. 1). Camera angle, distance and framing were varied across the set. Training used kohya_ss [bma25]; the configuration is listed in Table 1.

Parameter	Value	Parameter	Value
Dataset	150 renders	Optimizer	AdamW
Resolution	512×512	Schedule	cosine
Trigger token	*orgflw*	Batch / epochs / steps	1 / 10 / 3000
Rank / alpha	16 / 8	Training time	18 min
UNet LR	1×10^{-4}	Peak VRAM	4069 MiB
Text-encoder LR	5×10^{-5}	Checkpoint	epoch 9

Table 1: *Style adapter training configuration. VRAM peak occurred during sample generation. Training steps used 3106–3227 MiB.*

Since training loss plateaued from epoch 2 onward, checkpoint selection relied on visual comparison across epochs 2, 5, 7, and 9 (criteria in Section 3.5). Epochs 7 and 9 scored identically; epoch 9 was chosen for a closer material match too subtle for the five-point scale. It was used unchanged for all later runs (Fig. 2).

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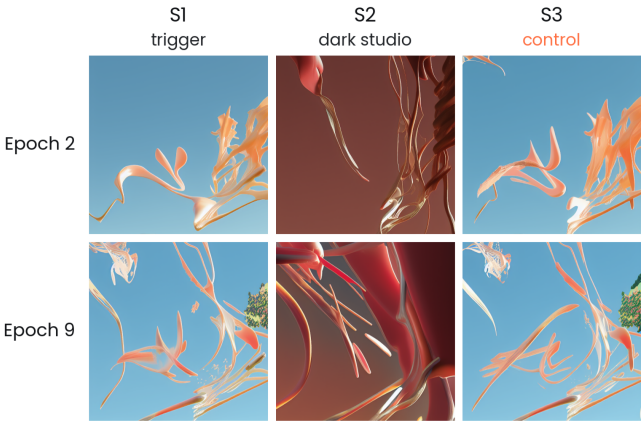


Figure 2: *Sample outputs at epochs 2 and 9 (seed 42, 20 steps, euler_a, guidance scale 7). Epochs 5 and 7 were also compared, not shown.* S1*: training caption with trigger token.* S2*: same, with a dark studio background replacing the sky.* S3*: control without the trigger token.*

3.3. Input Conditions

Eight input conditions were arranged as a cascade of decreasing similarity to the training data, on my assumption that style fidelity would decline along it: the original Cycles render (C01), low-poly version (C02), Eevee render (C03), Blender viewport solid shading (C04), flat-colour variant (C05), binary black-and-white mask (C06), an audio-reactive TouchDesigner structure following a tutorial by Tschepe [Tsc21] (C07), and greyscale noise (C08). All images were fed as still frames, so that each condition presented the same input in every run (Fig. 3).

3.4. Measurement Protocol

In Phase 1, all eight conditions were streamed with the style adapter active. Phase 2 repeated them without the adapter, isolating its contribution. Phase 3 reran C01 with the style adapter but only the first denoising step, isolating the second step's contribution. Phase 4 added one supplementary run in which the original network of input C07 was generated live rather than supplied as an exported still frame.

All other settings were fixed: seed 804281, prompt "orgflw, abstract organic shapes, blue sky background", step schedule "t_index_list [15, 45]", LCM-LoRA at 1.0, style adapter at 0.9, and resolution. The host machine remained plugged in, with identical open applications, untouched during runs except for timing and nvidia-smi readings.

Throughput was read from the component's server log. GPU memory, clock speed and temperature were logged via nvidia-smi. Each measurement window began 60 seconds after stream start, excluding warm-up. Memory is reported as total device allocation, an upper bound for pipeline consumption.

3.5. Rating the Outputs

Each run produced one output image, except for the supplementary run in Phase 4, where one frame was selected from the running stream. I rated the selected frames on five criteria (1–5 points each): organic morphology, colour palette and highlights, surface detail, overall stylistic coherence, and input preservation. The first four form the Style Index, with a possible range of 4–20. Input preservation is kept separate because it measures how much of the input's own structure survives, rather than how strongly the trained style comes through. To limit rater bias, the criteria were fixed before any output was examined closely. In addition, I rated all outputs myself in a single session, in randomised order, from neutrally named files. Only clear differences are interpreted as meaningful, since the scale has no defined unit and a one-point gap may reflect nothing more than where I set the boundary between two neighbouring grades.

4. Evaluation

4.1. Style Expression Across the Cascade

Contrary to the assumption stated in Section 3.3, these scores do not follow the original cascade order. Reducing the polygon count of the source geometry had no visible effect, C01 and C02 scored identically on all four criteria. The two inputs that retain the least information also scored high. A binary mask (C06) scored 13, two points above the fully shaded Eevee render (C03) at 11, while flat colour (C05) at 12 also outscored viewport solid (C04) at 8. The same ordering appears in the runs without the custom LoRA, though the spread there is only three points across all eight conditions, so it carries less weight (Fig. 4).

Two observations recur across the per-criterion scores. Conditions with a clearly separated figure, C01, C02, C05 and C06, occupy the upper half of the ranking, while those with low-contrast shading (C03, C04) or evenly distributed detail (C07, C08) occupy the lower half. C06 sits at the extreme of this pattern. Its figure-ground separation is the strongest in the series, and it is the only condition to reach 4 on both organic morphology and surface detail, the highest values observed. However, its colour palette score of 2 is the lowest among the well-performing conditions because its black background, the darkest in the series, remained black in the output (Fig. 3). The four conditions carrying colour scored 3 or 4 on colour palette, the four without scored 2.

The observations that follow rest on visual inspection of the eight styled outputs. They describe recurring patterns, and the explanations offered for them are hypotheses rather than tested mechanisms.

Where the input carried colour alongside a form close to the original render (C01, C02, C03, C05), both materials learned from the training data appeared in the positions they occupy in the renders. The organic forms were rendered in the salmon material, and the elements at the edge of the frame as green glitch structure.

Where the form remained close to the original but colour was absent (C04, C06), only one of the two appeared. No region was read as green glitch structure, and the whole figure was rendered in the salmon to pale pink material, carrying the organic form alone.

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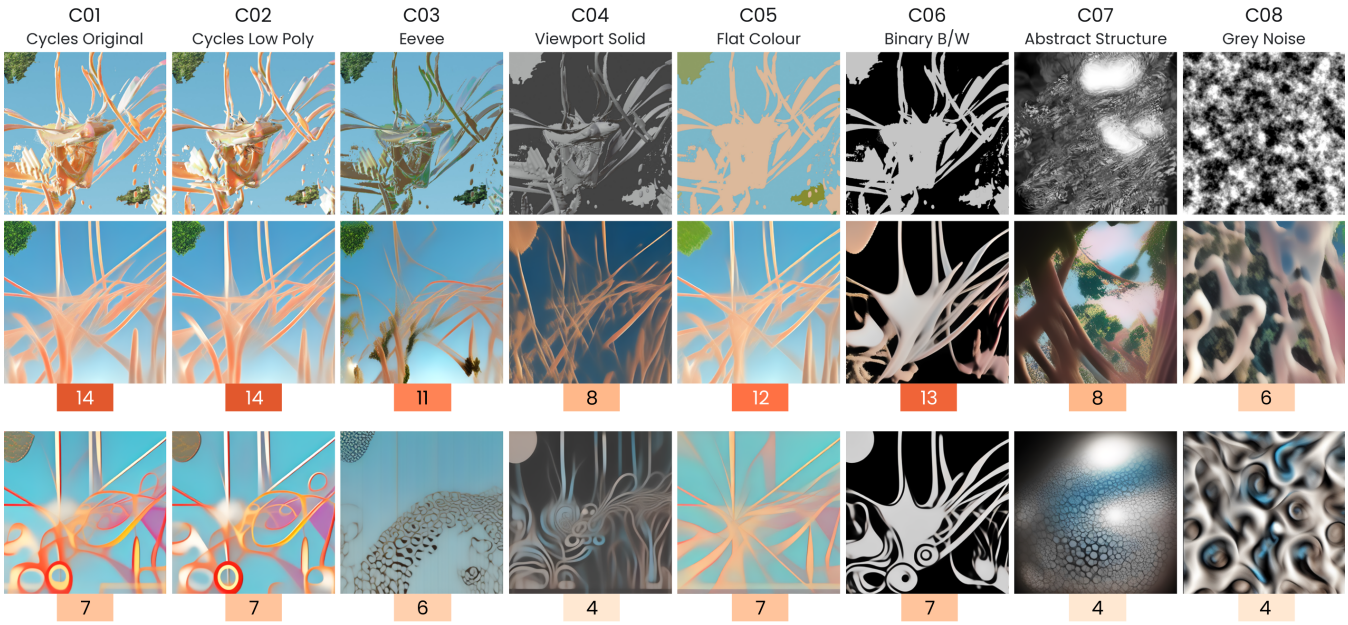


Figure 3: *The eight input conditions.* Top row*: input.* Middle row*: Phase 1–output with the style adapter.* Bottom row*: Phase 2–output without it. Each badge gives the Style Index of that run, range 4–20.*

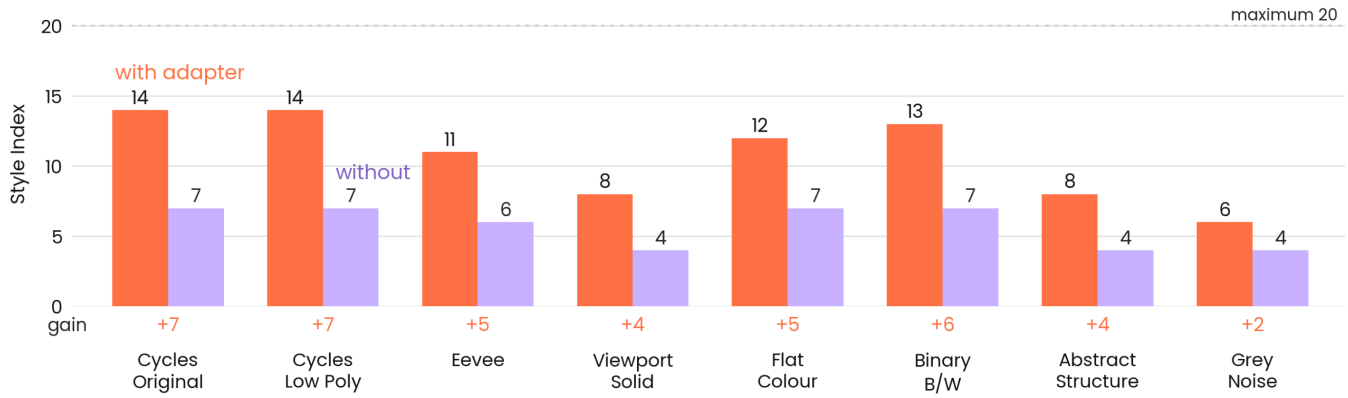


Figure 4: *Style Index for each input condition, with and without the style adapter. Sum of four criteria rated 1–5, range 4–20. The orange figure below each pair is the gain.*

C07 and C08 carried neither colour nor the original form. Their outputs showed a wider range of colour than C04 and C06, including green structure. The absence of green structure in C04 and C06 can therefore not be attributed to the absence of colour alone. What appeared to govern placement in these two conditions was local brightness, with darker regions filled by training elements that are themselves dark and brighter regions by lighter ones.

Two properties of the input appear to act separately on form. Which forms appear follows how close the input structure is to the training renders. How sharply those forms are defined follows the contrast in the input. C03 and C04 stay close to the original composition, but their strands have soft, washed-out edges. C07 and C08 produce forms that resemble the training renders far less as a whole, yet their strands have clear edges and remain distinct from one another. C06 is the only condition in which both hold at once, a composition close to the original and the sharpest separation in the series.

4.2. What the Adapter Contributes

The adapter raised the Style Index in every condition, from +7 for the two Cycles conditions down to +2 for greyscale noise (Fig. 4). The conditions that gained most are also the ones that scored highest with the adapter. Whether this reflects amplification of what the input carries is an interpretation, not a demonstrated mechanism.

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Input preservation stays the same or improves in six of the eight conditions and drops by one and two points in the binary mask and the audio-reactive structure, differences too small on a five-point scale to interpret.

The binary mask condition further demonstrates that colour is introduced by the adapter rather than inherited from the input. Although the input is strictly black and white, the styled output exhibits colour. In the matched run without the adapter, no colour was produced (Fig. 3).

Surface detail remains low throughout, reaching 4 only for the binary mask and otherwise sitting at 2 or 3. The small mesh fragments that surround the organic forms in the training data do not appear in any output. Such detail loss and surface smoothing were already visible in the training samples (Fig. 2), so they cannot be attributed to the input conditions alone.

4.3. Cost

Throughput showed no systematic dependence on either the input condition or the adapter. All sixteen runs of Phases 1 and 2 fell between 7.5 and 8.0 fps (Table 2). Memory never became the constraint — peak allocation was 4 439 MiB, leaving about 3.6 GB of the 8 GB unused. The card instead ran hot throughout, with mean temperatures of 86 to 87 °C, run maxima up to 94 °C (Table 2), and brief clock reductions to 1650 MHz on individual readings.

Removing the second denoising step raised throughput from 7.5 to 11.8 fps, a factor of 1.57, while the Style Index stayed at 14.

	fps	VRAM MiB	Load %	Clock MHz	Temp °C
P1, adapter	7.5–8.0	4174–4379	90–92	2161–2228	89–94
P2, no adapter	7.5–8.0	4159–4312	90–92	2183–2231	88–92
C01, two steps	7.5	4379	91	2228	92
C01, one step	11.8	4251	91	2284	89
C07, still frame	7.8	4235	92	2192	89
C07, live input	6.6	4241	92	2053	90

Table 2: *Throughput and GPU load. Values are means, temperature the maximum per run. The top block gives ranges across the eight runs of Phases 1 (P1) and 2 (P2), the rows below are single runs.*

4.4. Generating the Input on the Same GPU

In Phase 4, generating the C07 input live in TouchDesigner instead of supplying it as a still frame reduced throughput from 7.8 to 6.6 fps, a loss of about 15 %, while memory and GPU utilisation were unchanged (Table 2). The mean clock speed was 6 % lower in the live run, which may itself follow from the added load. The run was not rated visually, since its input is a different frame.

5. Discussion and Conclusion

5.1. What Determines Whether the Style Appears

The cascade did not predict the result. Since the noise level, and with it the bound on how far the output can depart from the input, was identical across conditions [MHS*22], the difference may instead depend on what remains after noising. Large, coherent, high-contrast structure appears to survive the noising step better than fine or low-contrast structure, which is my own conjecture rather than an established property. In C03 and C04, low contrast may cause boundaries to fade, producing washed-out edges. In C07 and C08, higher contrast seems to preserve sharper edges, even when the overall form no longer resembles the training data.

Based on this interpretation, the highest-scoring conditions combine the following attributes: a composition similar to the training renders, sufficient contrast to preserve distinct boundaries, and colour in the input.

A second pattern appeared at a different scale. Brightness does not order the conditions, but within an image the local value of each region appeared to determine which elements from the training data were placed there. As noted in Section 2, Akten et al. report this coupling in a conditional GAN, and a comparable pattern appears here in a few-step diffusion pipeline [AFG19].

The trigger token, by contrast, proved ineffective. When the adapter remained active but *orgflw* was omitted, control samples stayed nearly identical to outputs containing the token at every checkpoint (Fig. 2). Because all 150 captions were identical, the trigger never occurred separately from the subject description, leaving the model with no basis for distinguishing between the two. Koh et al. describe this form of entanglement [KMK*26]. In the streaming pipeline this has a practical consequence. The weight of the adapter can be reduced alongside the acceleration adapter [LTP*23], but the prompt cannot reliably turn on or off the style.

5.2. What Personalisation Costs

The style adapter added no measurable speed or memory cost. Its weights are merged before streaming, so the model that runs is the same size as the base model and performs the same operations, a property of merged low-rank adapters [HSW*22]. However, throughput depended on the denoising schedule and on the image source sharing the same GPU. For the motivating use case, the remaining GPU budget determines how complex that source can be.

5.3. Limitations

This study covers one artist, one style, one dataset of 150 images, and one GPU. All outputs were rated by the author from a single frame and seed per condition. Because the Style Index sums four equally weighted five-point criteria, it ranks conditions rather than measuring distance between them, and rating repeatability was not tested. Notably, the cascade is a designed ordering, not a computed measure of distance to the training distribution. Since the contrast and brightness explanations were developed after rating, from the same eight cases they describe, they fit these results by construction; testing them would require new images. Style was assessed only on still frames, which provides no information about moving input or temporal coherence. The low surface detail already appears in the training samples, preventing this study from separating training effects from pipeline effects. A single run of a single

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network was used for the live-input measurement. Without TensorRT, every throughput figure reported here is a lower bound. Of the two main results, the cost finding should transfer more readily, since it follows from adapter merging rather than from what the adapter learned. The brightness finding may instead be specific to this dataset: all 150 training images had bright backgrounds, leaving the model little dark content to place. An artist training on dark imagery would likely encounter the same gap in bright areas.

5.4. Conclusion

Returning to the question that motivated this work, on this hardware the input governed whether the style appeared, while the adapter added no measurable cost. Which forms appeared depended on how closely the input matched the training renders, while sharpness depended on contrast. Polygon count had no visible effect. A flat-colour reduction rendered without light transport landed two points below a path-traced render on the 4–20 scale, making it a viable substitute for the use case that motivated this work. The low-contrast conditions, meanwhile, scored lowest among those that kept the original composition. On the cost side, the fused adapter did not affect throughput or memory across sixteen runs; using one denoising step instead of two raised throughput by a factor of about 1.57 at an unchanged Style Index, while generating the input on the same GPU reduced throughput by approximately 15 %. Memory was never the binding constraint.

5.5. Future Work

Future work can test these findings more directly. One experiment would test colours outside the training palette with fixed geometry and luminance, to separate the effect of colour as such from that of a match to the trained palette. Two further tests would vary contrast at constant mean brightness and geometry. One would amplify the light and dark areas of the viewport-solid condition, and the other would soften the edges of the binary mask across several blur radii. Both test whether output sharpness follows edge contrast.

A second evaluator and blind re-rating would show how far the ratings depend on individual judgment. Moving input should also be tested, either as pre-rendered flat-colour animations switched during performance or as low-poly geometry processed live in TouchDesigner and reduced to a mask. Retraining with varied captions and backgrounds could make the trigger token usable as a control. Since production TensorRT support for this GPU generation is announced only from a release later than the one used here [Hin25], measuring the available performance headroom would require repeating the series on that version. Applying the same protocol to another style would show how far these findings carry beyond this single case, and whether real-time personalisation offers a practical path from offline rendering to live performance.

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